

HDIMS

Healthcare Dynamic Intelligence & Monitoring System — Project Abstract

PROBLEM STATEMENT India's Ayushman Bharat Digital Mission has linked over 900 million health accounts and generated more than 82 crore digital medical records, yet this data rarely reaches a clinician when it is clinically needed. Adoption research shows that patients, especially those elderly, rural, or with low digital literacy, cannot operate the apps and IDs this infrastructure depends on, leaving vast stores of health data clinically idle. HDIMS (Healthcare Dynamic Intelligence & Monitoring System) closes this gap with a clinician-facing intelligence layer that places zero operating burden on the patient.

OBJECTIVES Ingest whatever patient data already exists, whether linked digital records, local hospital systems, or live vitals, regardless of completeness; compute a continuously updated clinical risk state for each patient; automatically reconfigure which caregiver can access which data, at what depth, and for how long, as risk changes; and make every access change auditable in plain language.

METHODOLOGY A risk-state engine reads available clinical signals and drives a central access-orchestration engine, mapping each risk tier to a role-based access lattice spanning the assigned nurse, attending physician, on-call specialist, rapid-response team, referral hospital, and a designated family contact, with every grant expiring automatically as risk resolves. A parallel consent-and-audit layer records each change in readable language, while an escalation layer pre-provisions the right people before a crisis peaks rather than after.

INDUSTRIAL RELEVANCE HDIMS is built to sit inside resource-constrained public and private hospitals, integrating with existing hospital systems rather than replacing them, so it can be piloted and scaled without an infrastructure overhaul.

EXPECTED OUTCOME Faster, better-targeted response to patient deterioration, reduced alert fatigue through access-based rather than notification-based escalation, minimum-necessary data exposure by design, and a visible, auditable trust layer that strengthens confidence in digital health systems without demanding any digital literacy from the patient.